1

MATHEMATICS

Real Functions BOS

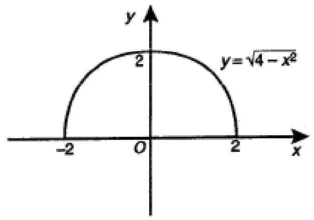
TEACHER'S COPY

1 2 3 4 5

Question 1

2002

Sketch the graph of $y=\sqrt{4-x^2}$ and state its domain and range.



Range is $0 \leqslant y \leqslant 2$.

Question 2

2001

State the domain and range of the function $y=2\sqrt{25-x^2}$

Domain: $-5 \le x \le 5$
Range: $0 \le y \le 10$

Question 3

1997

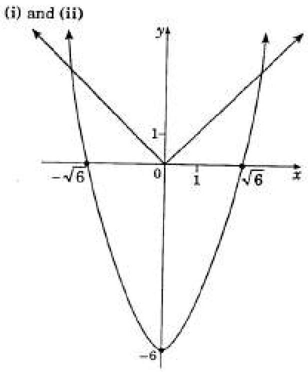
i) Sketch the graph of $y = x^2 - 6$, and label all intercepts with the axes

ii) On the same set of axes, carefully sketch the graph of y = |x|

iii) Find the x coordinates of the two points where the graphs intersect

iv) Hence solve the inequality $x^2 - 6 \le |x|$

(i) and (ii)



(iii) Solve $x^2 - 6 = x$ for $x > 0$.

$$x^2 - x - 6 = 0$$
$$(x-3)(x+2) = 0$$
$$\therefore\ x = 3 \text{ or } -2.$$

Reject $x = -2$ (outside domain).

The x coordinates of the two points where the graphs intersect are 3 and -3.

(N.B. Both graphs are even functions.)

(iv) We want the x values for which the parabola is 'on' or below the straight lines, that is $-3 \le x \le 3$.

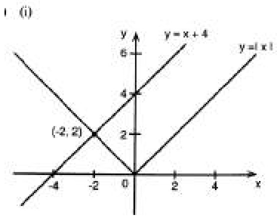
Question 4

1995

i) Draw the graphs of y = |x| and y = x + 4 on the same set of axes

ii) Find the coordinates of the point of intersection of these two graphs

(i)



$$y = -x$$
$$y = x + 4$$
$$\therefore -x = x + 4$$
$$-2x = 4$$
$$x = -2$$
$$\therefore y = 2$$

$\therefore$ The point of intersection is $(-2, 2)$.

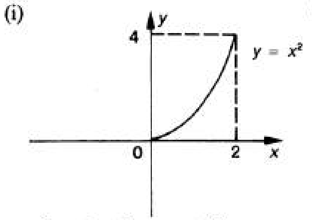
Question 5

1989

i) Sketch the function $y = x^2$, $0 \le x \le 2$, and write down its domain and range

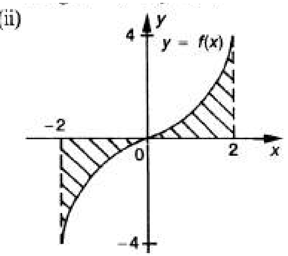
ii) If y in (i) is part of an odd function f(x) defined for $-2 \le x \le 2$, sketch on a second diagram the function f(x)

(i)



domain: $0 \leqslant x \leqslant 2$
range: $0 \leqslant y \leqslant 4$

(ii)



1

MATHEMATICS

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BOS

Name : ___________

1 2 3 4 5

Question 6

1978

For the function f(x) = √(1 - x^2)

a) state the natural (largest possible) domain

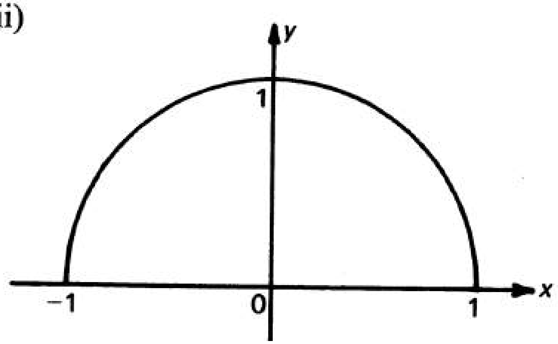
b) the range of the function

c) Sketch the graph of the function

(i) (a) $1 - x^2 \geqslant 0$
$(1 + x)(1 - x) \geqslant 0$
$\therefore\ -1 \leqslant x \leqslant 1.$

(b) $f(x) = \sqrt{1 - x^2}$
$f(0) = \sqrt{1} = 1$
$f(1) = f(-1) = \sqrt{1 - 1} = 0$
$\therefore\ 0 \leqslant f(x) \leqslant 1.$

(ii)



Question 7

1979

A function is defined by the following rule:

$$f(x) = \begin{cases} 0 & \text{if } x < -2, \\ -1 & \text{if } -2 < x < 0, \\ x & \text{if } x > 0. \end{cases}$$

Find:

a) f(-2) + f(-) + f(0)

b) f(a^2)

(a) $f(-2) + f(-1) + f(0)$
$= 0 + (-1) + 0$
$= -1.$

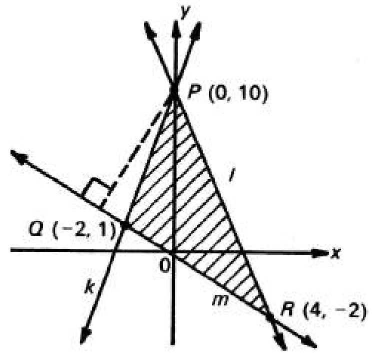
(b) $a^2 \geqslant 0$
Now $f(x) = x$ for $x \geqslant 0$
$\therefore\ f(a^2) = a^2.$

Question 8

1979

Show, by shading on a sketch, the region defined by the three inequalities

$$9x - 2y + 20 > 0,$$
$$3x + y - 10 < 0,$$
$$x + 2y > 0.$$



Question 9

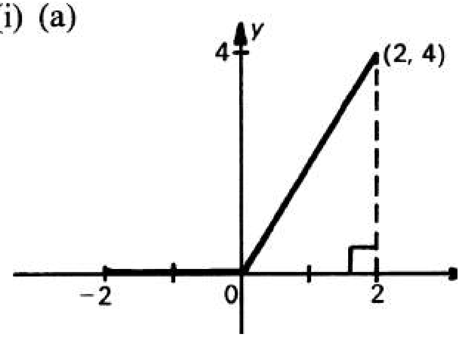
1980

The function f(x) is defined by the rule:

$$f(x) = \begin{cases} 0 \text{ if } x \leqslant 0, \\ 2x \text{ if } x > 0. \end{cases}$$

Sketch the function f(x) from x = -2 to x = 2

(i) (a)



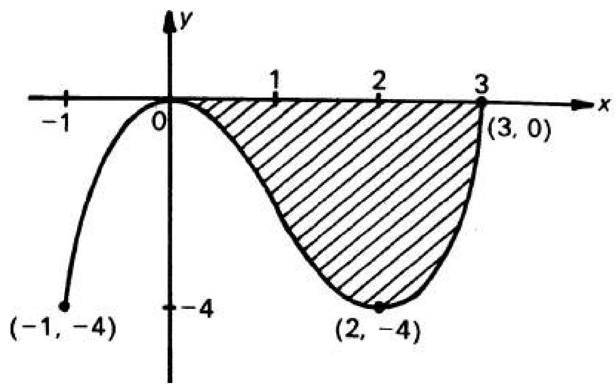
Question 10 **1980**

The function f(x) is defined by the rule

$f(x) = x^3 - 3x^2$ in the domain $-1 \le x \le 3$

a) Draw a sketch of the graph of f(x), clearly showing the x and y intercepts

b) Indicate on your sketch the region bounded entirely by the parts of the graph of y = f(x) and the x axis

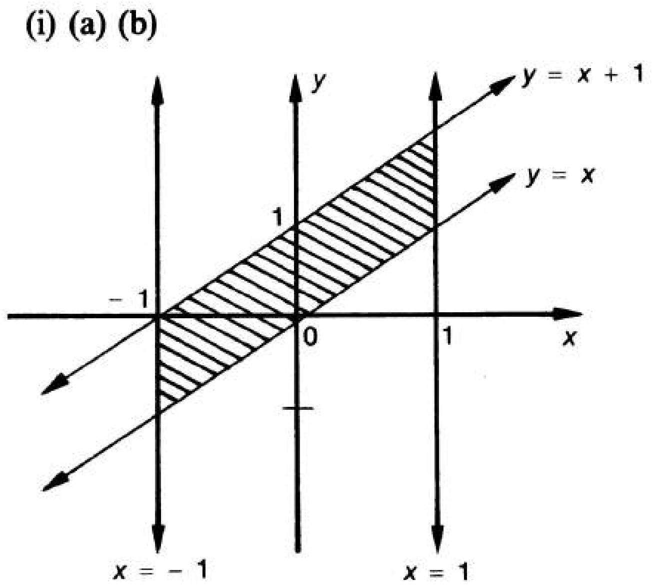


Question 11 **1985**

a) On the same set of axis, sketch the functions y = x and y = x + 1

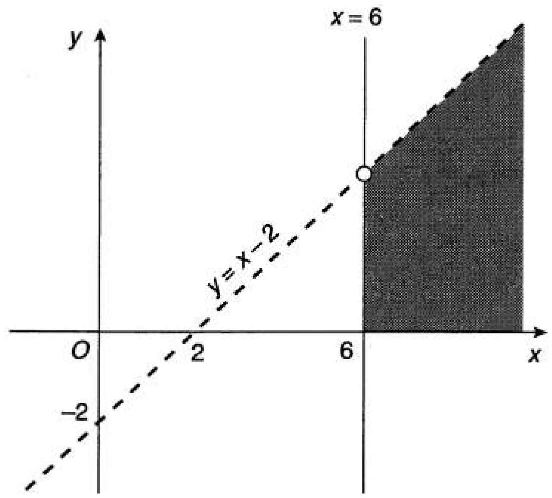
b) Indicate on your diagram, by shading the region which satisfy the following inequalities

$$|x| \le 1 \quad and \quad y \ge x \quad and \quad y \le x + 1$$



Question 12 **2003**

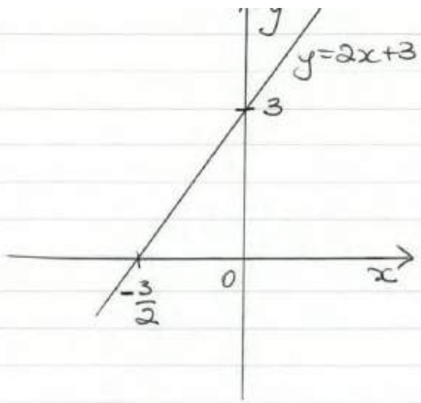
Shade the region in the Cartesian plane for which the inequalities $y < x - 2,\ y \ge 0$ and $x \ge 6$ hold simultaneously.



Question 13 **2009**

Sketch the graph of $y-2x=3$, showing the intercepts on both axes.

$y-2x=3$
$y=2x+3$



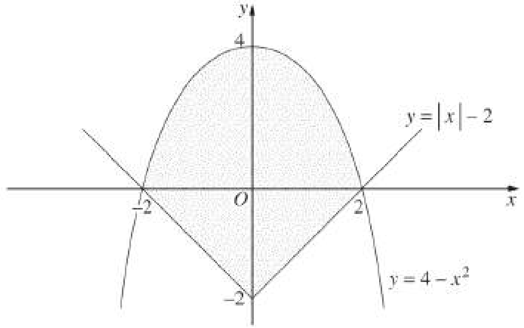
Question 14 **2010**

Let $f(x)=\sqrt{x-8}$. What is the domain of $f(x)$?

$f(x)=\sqrt{x-8}$, $x-8\geq 0$, $x\geq 8$, dom $f(x)$ is $[8,\infty)$

Question 15 **2011**

The diagram shows the graphs $y=|x|-2$ and $y=4-x^2$.



Write down inequalities that together describe the shaded region.

$y\leq 4-x^2$ and $y\geq |x|-2$

1

MATHEMATICS

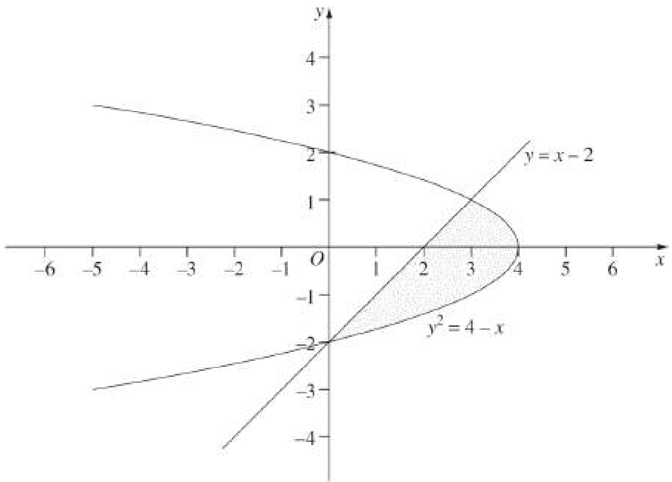
Real Functions
BOS

Name : ___________

1 2 3 4 5

Question 16 — 2012

The diagram shows the region enclosed by $y = x - 2$ and $y^2 = 4 - x$.



Which of the following pairs of inequalities describes the shaded region in the diagram?

(A) $\boxed{y^2 \le 4 - x \text{ and } y \le x - 2}$

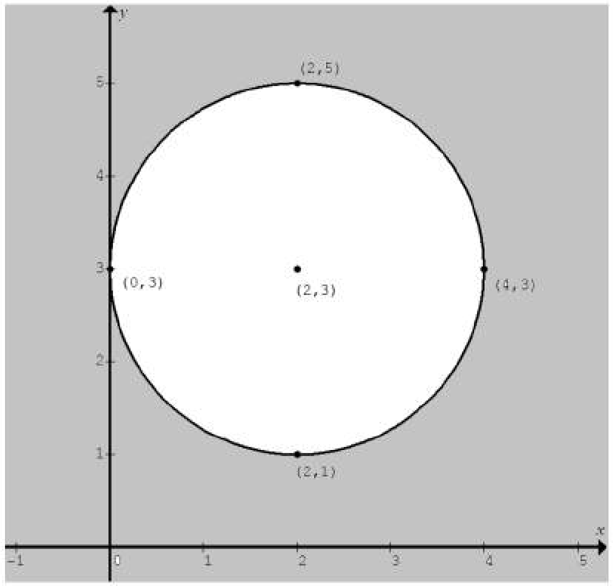
(B) $y^2 \le 4 - x$ and $y \ge x - 2$

(C) $y^2 \ge 4 - x$ and $y \le x - 2$

(D) $y^2 \ge 4 - x$ and $y \ge x - 2$

Question 17 — 2013

Sketch the region defined by $(x-2)^2 + (y-3)^2 \ge 4$.

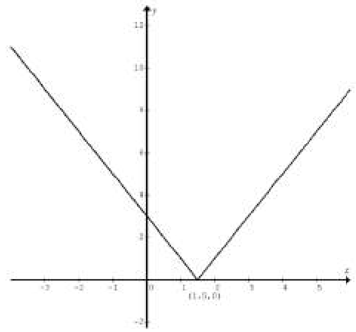


Question 18 — 2013

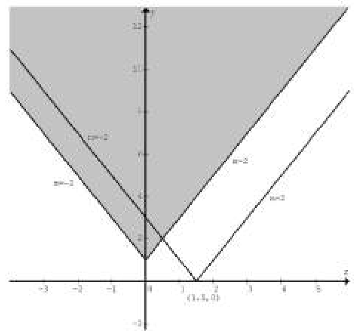
(i) Sketch the graph $y = |2x - 3|$.

(ii) Using the graph from part (i), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution.

Q15ci



Q15cii



To have exactly one solution, $y = mx + 1$ **either** passes through the shaded region (not including the boundary lines, $y = -2x + 1$ and $y = 2x + 1$), $\therefore\ m < -2$ or $m > 2$,

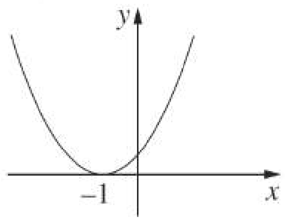
or through the point $\left(\frac{3}{2}, 0\right)$ and $\therefore\ m = -\frac{2}{3}$.

Question 19

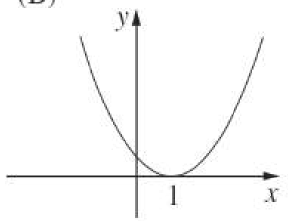
2014

Which graph best represents $y=(x-1)^2$?

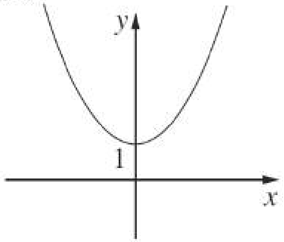
(A)



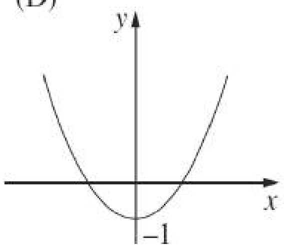
(B)



(C)



(D)

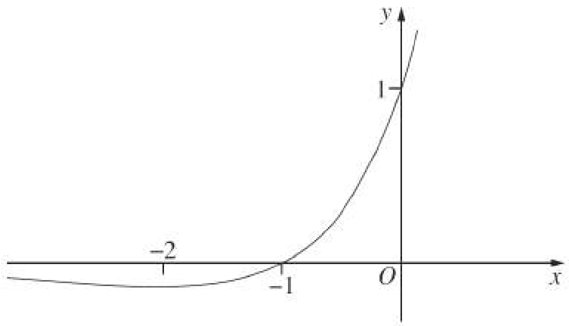


Q2 The graph of $y=(x-1)^2$ is the graph of $y=x^2$ translated to the right by 1 unit. B

Question 20

2015

The diagram shows the graph of $y=e^x(1+x)$.



How many solutions are there to the equation $e^x(1+x)=1-x^2$?

(A) 0

(B) 1

$\boxed{\text{(C)} \quad 2}$

(D) 3

Draw the parabola (with intercepts $(0,1)$, $(-1,0)$, $(1,0)$) over the graph. There are two intersections.

Answer: C

Question 21

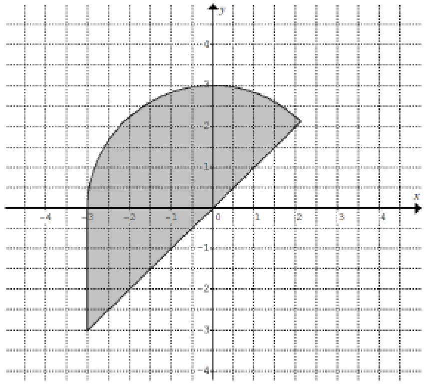
2015

(i) Find the domain and range for the function $f(x)=\sqrt{9-x^2}$.

(ii) On a number plane, shade the region where the points (x, y) satisfy both of the inequalities $y \le \sqrt{9-x^2}$ and $y \ge x$.

Q13bi $y=\sqrt{9-x^2}$ is a semi-circle of radius 3 units centred at the origin O in the first and second quadrants.
.: the domain is $[-3,3]$ and the range is $[0,3]$

Q13bii



1

MATHEMATICS

Real Functions
BOS

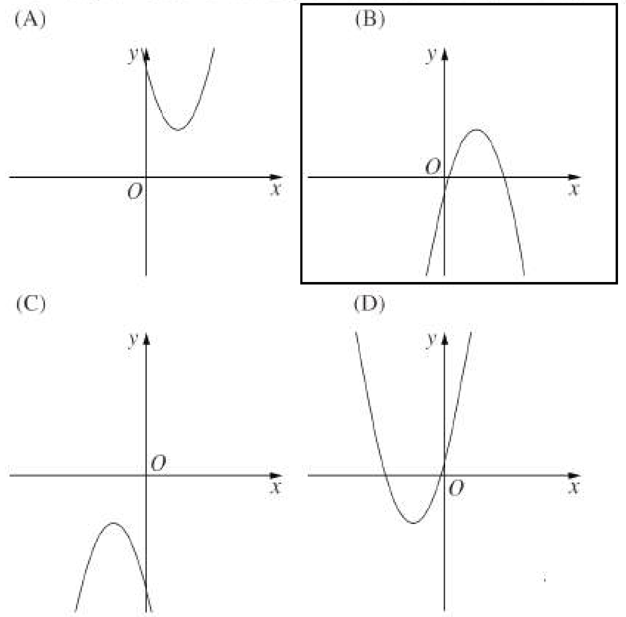
Name : ___________

1 2 3 4 5

Question 22

2016

Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?

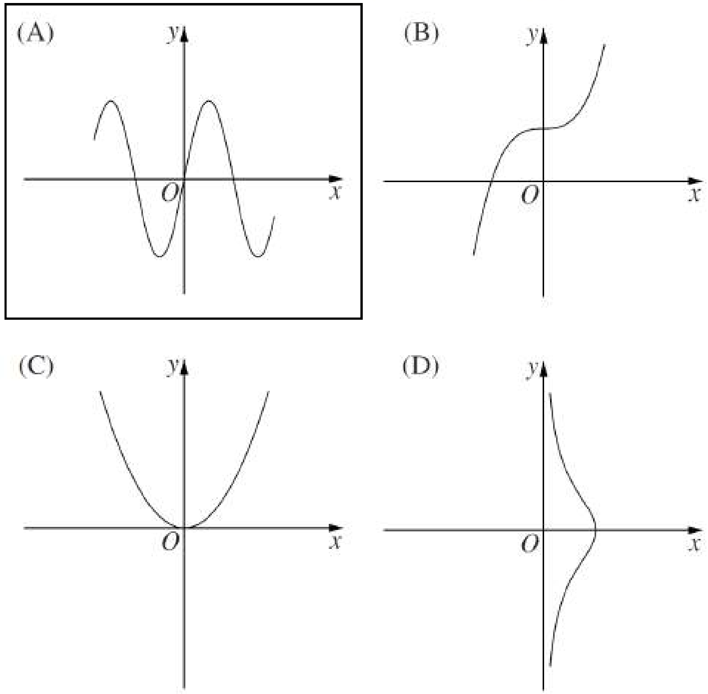


Inverted, turning point at $(2, 3)$

Question 23

2016

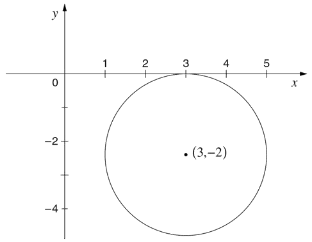
Which diagram shows the graph of an odd function?



Question 24

2016

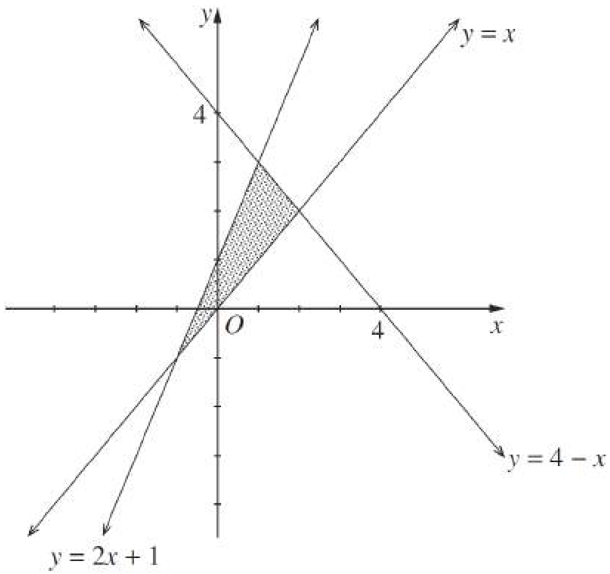
Sketch the graph of $(x-3)^2 + (y+2)^2 = 4$.



Question 25

2017

The region enclosed by $y = 4 - x$, $y = x$ and $y = 2x + 1$ is shaded in the diagram below.



Which of the following defines the shaded region?

A. $y \le 2x+1,\quad y \le 4-x,\quad y \ge x$

B. $y \ge 2x+1,\quad y \le 4-x,\quad y \ge x$

C. $y \le 2x+1,\quad y \ge 4-x,\quad y \ge x$

D. $y \ge 2x+1,\quad y \ge 4-x,\quad y \ge x$

Question 26 **2017**

Find the domain of the function $f(x)=\sqrt{3-x}$.

$f(x)=\sqrt{3-x}$

$3-x\geq 0$

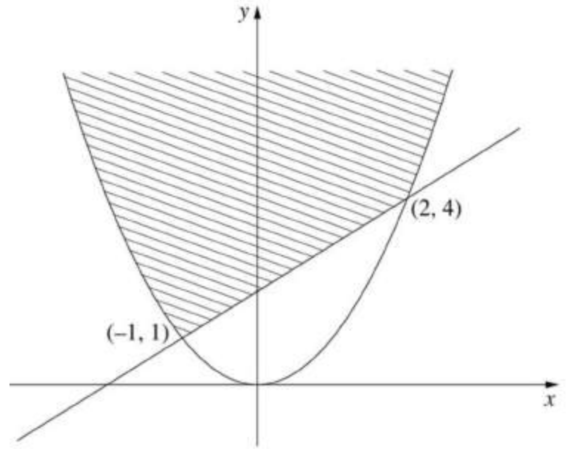
$3\geq x$

$\therefore \quad x\leq 3$

Question 27 **2019**

The parabola $y=x^2$ meets the line $y=x+2$ at the points $(-1, 1)$ and $(2, 4)$. Do NOT prove this.

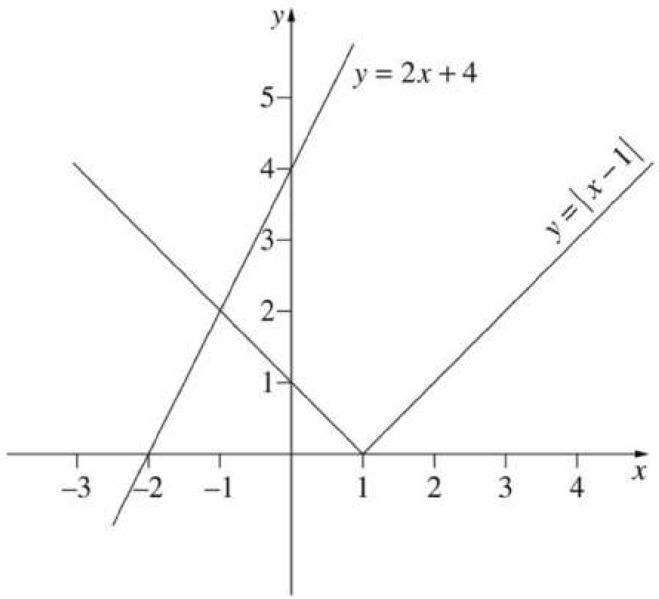
By first sketching the graphs of $y=x^2$ and $y=x+2$, shade the region which simultaneously satisfies the two inequalities $y\geq x^2$ and $y\geq x+2$.



Question 28 **2019**

(i) Sketch the graph of $y=|x-1|$ for $-4\leq x\leq 4$.

(ii) Using the sketch from part (i), or otherwise, solve $|x-1|=2x+4$.



$$2x+4=-(x-1)$$
$$2x+4=-x+1$$
$$3x=-3$$
$$x=-1$$